



Course Syllabus ELPT 2319 H03A Programmable Logic Controllers I

Instructor/Department Information

Instructor: David Loveless

Email: dloveless@lee.edu

Office Location: No Office

Phone: No Office Phone

Division Chair: Waldo Collins

Division Chair Contact Information: wacollins@lee.edu; 281/425-6474; TV1-214D

Administrative Office Specialist: Iris Gonzalez

A.O.S Contact Information: igonzalez@lee.edu; 281/425-6289; TV1-214B

Executive Director: Industrial & Technical Programs: Dr. Richard Davis

Vice President: Academic Affairs: Dr. Janena Norris

Security Information: Eric Williams, Security Chief; ewilliams@lee.edu; 281/425-6888

Course Details

Fall 2026

Session: Regular Academic Session

Location: TV1208

Day(s): Monday Wednesday

Times: 07:30 PM-08:20 PM

Lab Location: TV1208

Lab Day(s): Monday Wednesday

Lab Time(s): 08:25 PM-09:15 PM

Course Description

Fundamental concepts of programmable logic controllers, principles of operation, and numbering systems as applied to electrical controls. Lecture Hrs = 2, Lab Hrs = 2

Credit Hours: 3

Student Learning Outcomes

1. Identify and describe digital logic circuits and explain numbering systems; Explain the operation of programmable logic controllers; Convert ladder diagrams into programs; Incorporate timers and counters utilizing programmable logic controllers;
2. Execute and evaluate programs.

Prerequisites/Co-requisites

ELPT 1411 PreReq

Student/Office Hours

The official communication platform for this class will be via blackboard announcements and blackboard messages. Students are encouraged to schedule a meeting with the course instructor for any questions or concerns regarding the course. In order to schedule a meeting, please contact the course instructor using the blackboard or the email listed above. Please allow 2 business days for a response. See instructor office hours below which are by appointment only. Simply contact me so we can decide when to meet in person for concerns that can not be addressed using Blackboard messages or email.

Student/Office Hours

Day	Time
Monday	6:30-7:30 PM
Wednesday	6:30-7:30 PM

Required Textbook/Supplies Requirement

Title: Programmable Logic Controllers

ISBN: 9781264163342

Authors: Frank Petruzella

Publisher: McGraw Hill

Edition: 6

required/optional: Required

Material Type: ETEXTBOOK

myBooks Information

Non-dual-credit students are automatically enrolled in [myBooks](https://www.lee.edu/mybooks/), which provides digital and paper textbooks and other course materials. The cost of myBooks is \$33.00 per semester credit hour and applies to courses that have a required textbook or course kit. Students who wish to purchase their own books can opt out of myBooks by selecting the opt-out link in myLC Campus. If you need assistance, contact the Instructional Materials Manager at myBooks@lee.edu. For more information, please visit <https://www.lee.edu/mybooks/>.

Dual Credit students are not automatically opted in to myBooks. Please meet with your Lee College Dual Credit Advisor to learn more about the textbook process. You can find your dual credit advisor by visiting <https://www.lee.edu/dual/meet/>.

Technology

In our efforts to prepare students for a changing world, students may be expected to utilize computer technology in this course. Students may utilize available computer applications by logging into their One Login account located on the [Lee College Login webpage](#).

All emails from the college are sent to your myLC email account. Directions for forwarding your MyLC email to your personal email address are listed on the MyLC email page; however, emails from the college may include private educational records or other private information and the college is not responsible if this information is forwarded to an insecure personal email account.

Schedule/Outline

The first table shows opening dates for each unit in the course. The second table lists assignments and due dates.

DATE	UNIT
Monday, August 24	00 - BASIC COURSE TOOLS UNIT
Monday, August 24	0 - OVERVIEW
Monday, August 24	1 - COURSE INTRODUCTION
Monday, August 24	2 - AN INTRODUCTION TO PLCs
Monday, August 31	3 - SWITCHES AND LOADS
Monday, September 7	4 - RELAY TECHNOLOGY
Monday, September 14	5 - LATCHING RUNG

DATE	UNIT
Monday, September 21	6 - MOTOR CONTROL
Monday, September 28	7 - STAGED MOTORS
Monday, October 5	8 - TURN ON LIGHTS
Monday, October 12	9 -TRAVELING LIGHTS
Monday, October 19	10 - TRAFFIC LIGHT
Monday, October 26	11 -CHEAP CAR WASH
Monday, November 2	12 -CHEAP AND ULTRA CAR WASH
Monday, November 9	13 -COMPLETE CAR WASH
Monday, November 16	14 -SOFT DRINK FACTORY CANNING
Monday, November 23	15 - SOFT DRINK FACTORY FLAVOR
Monday, November 30	16 -COMPLETE SOFT DRINK FACTORY
Monday, December 7	FINAL EXAM

Assignment Type	Assignment Name	Due Date
ASSIGNMENT	Upload an Image of Your PLC PROJECTS sub-folders in your OneDrive ELPT 2319 Folder	8/31/26, 12:00 AM
QUIZ	Getting Started in this Course	8/31/26, 12:00 AM
QUIZ	Course Introduction and Expectations	8/31/26, 12:00 AM
QUIZ	Syllabus Comprehension	8/31/26, 12:00 AM
QUIZ	ELPT 2319 Directory Assignment	8/31/26, 12:00 AM
EXAM	01: Chapter 1 - PLC Basics: Role of PLCs in industry PLC vs relay control PLC hardware overview	8/31/26, 12:00 AM
QUIZ	Overview of PLCs Video	8/31/26, 12:00 AM
QUIZ	AutomationDirect PLC Handbook CH 1-2	8/31/26, 12:00 AM
ASSIGNMENT	Discrete Input Measurement Report Upload Center	9/7/26, 12:00 AM
EXAM	02: Chapter 9 - Discrete Process Measurement (Kuphaldt)	9/7/26, 12:00 AM
QUIZ	Chapter 22 - Relays (Petruszella)	9/14/26, 12:00 AM
EXAM	03: Chapter 22 - Relays (Petruszella)	9/14/26, 12:00 AM
QUIZ	Digital Challenge Intro and Binary Counting	9/21/26, 12:00 AM

Assignment Type	Assignment Name	Due Date
QUIZ	First PLC Projects Assignments Info Sheet	9/21/26, 12:00 AM
EXAM	04: Chapter 2 - Number Systems & Codes: Binary, BCD, hexadecimal conversions and data representation	9/21/26, 12:00 AM
PROJECT	Latching Rung Project Upload Center	9/21/26, 12:00 AM
DIGITAL CHALLENGE	01: Binary Counting Upload Center	9/21/26, 12:00 AM
DIGITAL CHALLENGE	02: Octal Counting Upload Center	9/21/26, 12:00 AM
DIGITAL CHALLENGE	03: Hexadecimal Counting Upload Center	9/21/26, 12:00 AM
QUIZ	Binary to Decimal - Convert Numbers and Fractions Like a Pro!	9/28/26, 12:00 AM
EXAM	05: Chapter 3 - Logic Concepts: Boolean logic AND/OR/NOT functions	9/28/26, 12:00 AM
PROJECT	Motor Control Project Upload Center	9/28/26, 12:00 AM
DIGITAL CHALLENGE	04: Number Conversions One Upload Center	9/28/26, 12:00 AM
DIGITAL CHALLENGE	05: Number Conversions Two Upload Center	9/28/26, 12:00 AM
DIGITAL CHALLENGE	06: Binary Addition Upload Center	9/28/26, 12:00 AM
EXAM	06: Chapter 4 - Ladder Logic Fundamentals: Ladder diagrams, contacts, coils, seals, interlocks, and program scan cycle	10/5/26, 12:00 AM
PROJECT	Staged Motors Project Upload Center	10/5/26, 12:00 AM
DIGITAL CHALLENGE	07: Binary Subtraction Upload Center	10/5/26, 12:00 AM
DIGITAL CHALLENGE	08: Logic Gates One Upload Center	10/5/26, 12:00 AM
EXAM	07: Midterm	10/12/26, 12:00 AM
PROJECT	Turn On Lights Project Upload Center	10/12/26, 12:00 AM

Assignment Type	Assignment Name	Due Date
DIGITAL CHALLENGE	09: Logic Gates Two Upload Center	10/12/26, 12:00 AM
DIGITAL CHALLENGE	10: Name That Gate Upload Center	10/12/26, 12:00 AM
PROJECT	Traveling Lights Project Upload Center	10/19/26, 12:00 AM
EXAM	08: Chapter 5 - PLC Memory & Addressing: I/O addressing, data files, and registers	10/19/26, 12:00 AM
DIGITAL CHALLENGE	11: Name That Gate Two Upload Center	10/19/26, 12:00 AM
DIGITAL CHALLENGE	12: Truth Tables Upload Center	10/19/26, 12:00 AM
PROJECT	Traffic Light Project Upload Center	10/26/26, 12:00 AM
EXAM	09: Chapter 6 - Input/Output Devices: Discrete inputs and outputs, field wiring practices	10/26/26, 12:00 AM
PROJECT	Cheap Car Wash Project Upload Center	11/2/26, 12:00 AM
EXAM	10: Chapter 7 - Timer Instructions: On-delay, off-delay, retentive timers, and timing applications	11/2/26, 12:00 AM
PROJECT	Ultra and Cheap Car Wash Project Upload Center	11/9/26, 12:00 AM
EXAM	11: Chapter 8 - Counter Instructions: Up/down counters and industrial counting applications	11/9/26, 12:00 AM
PROJECT	Complete Car Wash Project Upload Center	11/16/26, 12:00 AM
PROJECT	Soft Drink Canning Project Upload Center	11/23/26, 12:00 AM
PROJECT	Soft Drink Flavor Project Upload Center	11/30/26, 12:00 AM
PROJECT	Soft Drink Factory Project Upload Center	12/7/26, 12:00 AM
FINAL EXAM	Exam 12 - Final Part A	12/7/26, 12:00 AM
FINAL EXAM	Exam 12 - Final Part B	12/7/26, 12:00 AM

Late Work Policy

There is no penalty for late work

Grading Components

Grading		
Assignment Types	How many of each Assignment (% Total per Assignment)	Percentage of total grade per Assignment type
Digital Challenge	12 assignments at (1.6667% of Total grade Each)	20%
Project Grades	12 assignments at (2.5% of Total grade Each)	30%
Exam Grades	11 assignments at (2.7273% of Total grade Each)	30%
Quiz	11 assignments at (0.4545% of Total grade Each)	5%
Final Exam	1 assignment at (10% of Total grade Each)	10%
Paper	1 assignment at (5% of Total grade Each)	5%
	Total - 48 assignments at (100% of Total Grade)	Total - 100%

*Faculty reserves the right to change a grading component and will provide written notice via email and announcements in Blackboard should a grading component substantively change.

Grading Scale

Grading	
Final Average in Percent	Letter Grade
90 - 100	A
80 - 89.49	B
70 - 79.49	C
60 - 69.49	D
< 60	F

Final Average in Percent	Letter Grade
Incomplete - no credit (See details below)	I

<http://catalog.lee.edu/content.php?catoid=3&navoid=70#grades>

Attendance Policy

<http://www.lee.edu/attendance-pol/>

Please note the Lee College attendance policy indicated in the link and notify me by phone or via email when you are not able to attend class. For the online attendance policy, in conjunction with the U.S. department of Education please follow the online attendance policy link above to review the policy.

Absences Due to Religious Holy Days

<http://www.lee.edu/absent-h-days/>

Students may be absent from classes for the observance of a religious holy day, provided they have given prior notice a week in advance and will be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time as established by the instructor.

Withdrawal / Drop Policy

Students are responsible for dropping classes. State-mandated refund policies, as well as drop deadlines, appear in schedules, and on the [college website](#).

It is recommended students contact their advisor before they withdraw and drop a class.

Students who drop their courses after the official day of record and prior to the final day for Student Drops may be subject to the [Six Drop Rule](#).

[Resignation \(Complete Withdrawal\):](#)

Students may resign from all of their classes in any semester prior to the start of finals week. Students are required to sign a statement indicating they understand impacts of the resignation and are encouraged to visit with an advisor if they have concerns. A resignation results in a grade of W4 for all courses.

Incomplete

<http://www.lee.edu/i-grades/>

A grade of "I" indicates incomplete work resulting from illness or other unavoidable circumstances. To be eligible to receive an "I" students must have completed at least 75 percent of the work required for the course in question during the original term of enrollment. To receive an "I," a student must enter into a contract with the instructor of the course regarding the work that is to be completed and the grade the student will receive in the event that the work is not completed. The remaining work must be completed within one semester unless extenuating circumstances require a longer period for completion. Instructors have the right to submit any grade at any time to replace an "I" grade, with a grade of "F." Students who receive "I" grades should not re-enroll for the class unless they are terminating the incomplete agreement and wish to start over with a new section of the class. In this case, the student is urged to contact the original instructor to request release from the incomplete agreement.

After the next long semester has lapsed, if the "I" grade has not been changed to another grade by the instructor, the "I" grade will be replaced with an "F".

Incompletes (noted by the grade "I") are not issued in developmental courses.

Course-Specific Policies

If you are enrolled in a traditional face-to-face course, ignore the following.

If you are enrolled in a HYBRID or an ONLINE class, please read carefully the following:
HYBRID COURSES

This hybrid class requires you to complete half of the coursework on your own time using Blackboard. The other half of the course will be conducted through routine face-to-face meetings. It is essential to stay on schedule, as hybrid classes can be more demanding than traditional face-to-face courses.

If your hybrid class includes a lab component, the labs will be completed during face-to-face sessions, while the lecture component will be conducted online.

If you have any questions about this course, please don't hesitate to reach out. Do not wait until you fall significantly behind.

For issues logging into Blackboard, contact the Lee College Help Desk at helpdesk@lee.edu or by phone at (281) 425-6952.

ONLINE COURSES

This online class requires you to complete all coursework independently using Blackboard. There will be no routine face-to-face meetings for this course.

It is essential to stay on schedule, as online classes can be more demanding than traditional face-to-face courses. Please note that online classes in Technical Studies do not include a lab component.

If you have any questions about this course, don't hesitate to reach out. Do not wait until you fall significantly behind.

For assistance with logging into Blackboard, contact the Lee College Help Desk at helpdesk@lee.edu or by phone at (281) 425-6952.

Academic Honesty Policy

<http://www.lee.edu/aca-honesty/>

Lee College expects and encourages all students to contribute to such an environment by observing the principles of academic honesty outlined in the College's Academic Honesty Code, which is detailed in Chapter Three of the Lee College Catalog under Academic Honesty. It is the student's responsibility to understand and comply with this code.

AI Policy

AI Encouraged with Attribution, Verification and Reflection: Using AI to find information, brainstorm ideas and test arguments is encouraged and expected. Copying and/or paraphrasing from an AI assistant is permitted as long as copied material is quoted or in a different colored font with a citation, and paraphrased material has a citation. Citations must include verification. To ensure a proper citation, follow these guidelines:

- The citation should indicate that the material was copied or paraphrased from an AI assistant.
- If the AI provided facts, sources and/or quotes, the citation should identify the facts, sources and/or quotes and confirm that they were independently verified and that they are correct and/or real.

In addition to copying and paraphrasing, you can present ideas and arguments provided by the AI assistant just as you can present ideas and arguments provided by other people; however, like other people, AI assistants can be biased even when the

facts, quotes and sources are real. Therefore, when using ideas from an AI assistant, you are required to reflect on the ideas and consider biases that may be present.

Reason: AI has become part of the network of people and technology that we use to gather information and ideas. Using AI is encouraged because it is important to understand how to exercise control over AI and engage thoughtfully with it. This includes verifying that facts, quotes, and sources are real, and that ideas and themes presented by the AI assistant are susceptible to bias.

General AI Principles:

- AI should help you think. Not think for you.
- You are responsible for the accuracy of facts, quotes, sources etc. provided by your AI assistant.
- Do not enter information into AI tools that you would not post on social media.
- You must disclose when you use generative AI in your work.
- The AI policy in this course may be different from the policy in other courses.

DEI Policy

Effective January 1, 2024, Senate Bill 17 (SB 17) prohibits community colleges from engaging in certain diversity, equity, and inclusion (DEI) initiatives, with some exceptions. One of the exceptions is that SB 17's prohibitions do not apply to academic courses; scholarly research or creative works by students, faculty, or research personnel or the dissemination of the research or works; or data collection. This means that faculty can include DEI topics in class discussion, readings, research projects, homework assignments, and other aspects of instruction, and students and faculty can exhibit and present their work without restrictions.

Non-Discrimination:

The College District prohibits discrimination, including harassment, against any student on the basis of an individual's Protected Characteristics including race, color, national origin, disability, religion, age, limited English proficiency or English learner status, veteran status, genetic information, sex (including pregnancy, parental status, sex stereotyping or treating people differently because they do not conform to sex-role expectations, sexual orientation, gender identity or gender expression) or any other basis prohibited by law. Retaliation against anyone involved in the complaint process is a violation of College District policy and is prohibited.

Complaints/Grievances

<http://catalog.lee.edu/content.php?catoid=27&navoid=750&hl=appeals#student-complaints>

Students are encouraged to contact their instructor for assistance regarding any aspect of the course. For issues that are not resolved after consultation with the instructor, students may contact the Division Chair. If an issue is not resolved after meeting with the instructor and division chair, the student can submit a formal complaint online by following the procedures in the link above.

Student Support and Services

<https://www.lee.edu/student-services/>

Lee College is proud to offer a variety of student services to provide a variety of supports within our division to assist students navigate their time at the college. You can find more information about student services at the link above.

Timely Care

[TimelyCare](#) is available to current credit students for free, providing virtual care to include mental health support, medical visits, and general well-being information. Download the app today by going to www.timelycare.com/lee

Access Center for Students with Disabilities

The Lee College [Access Center for Students with Disabilities](#) is located in Rundell Hall Office 101 and is the campus Disability Services unit that works with students who have disabilities to provide and/or arrange reasonable accommodations for students in their courses. Students who have, or think they may have, a disability, are invited to contact the Access Center for a confidential discussion at (832)-556-4069 or at kvillanueva@lee.edu.

[Link to the Americans with Disabilities Act \(ADA\) Website](#)

kNOw More (Title IX)

<https://www.lee.edu/titleix/know-more-campaign/>

Lee College is committed to providing all with a learning and work environment that is free from sexual harassment and assault. If you or someone you know has been harassed or assaulted, you can find the appropriate resources, both on and off campus at the [Title IX Available Resources Page](#). Confidential support can be found through The Bridge Over Troubled Waters. The Bridge can be contacted at 281-420-5600 (Baytown Office) or through their 24-hour hotline at 713-473-2801. Here is the link for the U.S. Department of Education – [Title IX – Sex Discrimination](#).

Lee College faculty are “responsible employees”, which means, as a faculty member I am required to report incidents of sexual misconduct and thus cannot guarantee confidentiality. I must provide the Title IX Coordinator with relevant details such as the names of those involved in the incident. If you want to make a formal report, you can contact Jose Martinez, Title IX Coordinator, at josmartinez@lee.edu or 281-425-6546 or email TitleIXSupport@lee.edu.